

# MAYILRAJ R *Software Engineer*

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Tuticorin, Tamil Nadu, India

## EXPERIENCE

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| <b>Ibee Analytics, Software Development Engineer Intern</b> <ul style="list-style-type: none"><li>Designed Python backend modules using OOP/OOAD principles and wrote pytest unit tests achieving &gt;90% regression coverage across 5+ API endpoints, reducing post-release bug discovery by ~30%.</li><li>Built and shipped 3 RESTful APIs with Node.js/Express.js following modular MVC architecture; performed impact analysis before each sprint release, cutting unplanned rollbacks to zero.</li><li>Developed React.js analytics dashboards integrated with Chart.js, visualising real-time KPIs for business stakeholders; optimised MongoDB schemas (compound indexes, projection) reducing average read latency by ~25%.</li><li>Collaborated across 4 Agile sprints using Git branching and PR-based code reviews, maintaining 100% on-time milestone delivery.</li></ul> | 08/2025 – 09/2025<br>Hyderabad  |
| <b>Hapus Infotech, Web Developer Intern</b> <ul style="list-style-type: none"><li>Refactored legacy Python/Django codebase by applying OOP design patterns (Factory, Strategy), cutting average page-load time by ~20% and reducing lines of duplicated code by 35%.</li><li>Implemented JWT-based user authentication and session management with input validation; designed responsive UI using Bootstrap 5, raising mobile usability score from 62 → 87 (Lighthouse).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                    | 07/2025 – 08/2025<br>Coimbatore |

## EDUCATION

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| <b>B.Tech – Information Technology, University College of Engineering, Tiruchirappalli</b><br>CGPA: 7.2 / 10                  | 08/2023 – 05/2026 |
| <b>Diploma – Electronics &amp; Communication Engineering, Thevanesam Eruthaya Ammal Polytechnic College</b><br>CGPA: 9.1 / 10 | 06/2020 – 04/2022 |

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, Embedded C, SQL | **Backend / APIs:** Node.js, Express.js, Flask, Django, REST API Design | **Frontend:** React.js, HTML5, CSS3, Bootstrap 5, Chart.js data visualisation | **Databases:** MongoDB, MySQL, PostgreSQL | **Tools & Practices:** Git/GitHub, Postman, VS Code, Antigravity, NPM, Agile/Scrum, OOP/OOAD, CI/CD pipelines

## CERTIFICATIONS & ACHIEVEMENTS

**Python Programming – GUVI (HCL) | Java Training – LinkedIn Learning | SQL Bootcamp – Lets Upgrade |**  
**Deloitte Cyber Job Simulation (Forage):** practised threat analysis, SIEM triage, and incident reporting workflows. | **TATA Cybersecurity Analyst Simulation (Forage):** completed IAM, network defence, and vulnerability assessment modules. | **MeitY – ISEA Digital Security Quiz:** Ministry of Electronics & IT, Government of India.

## PROJECTS

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| <b>Predictive Marketing Dashboard, Python, Flask, Facebook Prophet, BeautifulSoup, REST API</b> <ul style="list-style-type: none"><li>Architected a modular Flask backend using OOAD patterns (Repository, Service Layer), enabling independent unit testing of each component with pytest; achieved 95%+ test coverage on the forecasting pipeline.</li><li>Integrated Facebook Prophet time-series model to forecast marketing KPIs with &lt;8% MAPE on validation data; automated SEO metric collection via BeautifulSoup scrapers exposed as REST endpoints.</li><li>Delivered iterative milestones with documented impact analysis; the dashboard reduced manual reporting effort by an estimated 4 hours/week for the client team.</li></ul> |
| <b>Empower Her – Safe Route Navigation, Android Studio, Google Maps SDK, Firebase, AI/ML</b> <ul style="list-style-type: none"><li>Designed OOP architecture for an Android safety app with AI-based threat detection module; wrote unit tests for GPS parsing and real-time alert logic, achieving zero false-negative alerts in testing.</li><li>Integrated Firebase Realtime Database for &lt;200 ms GPS-sharing latency; implemented smart-ring SOS trigger via BLE, enabling one-tap emergency broadcast to 5 pre-configured contacts.</li></ul>                                                                                                                                                                                              |
| <b>Student Management System, Python/Django, JavaScript, Bootstrap 5, MySQL</b> <ul style="list-style-type: none"><li>Built a full-stack MVC system following OOAD class-diagram design; implemented role-based access control (Admin, Faculty, Student) with Django's permission framework, securing 3 distinct data scopes.</li><li>Wrote unit and integration tests for enrolment, attendance, and authentication modules using Django TestCase; achieved zero regressions across 12 feature releases during development.</li></ul>                                                                                                                                                                                                             |